4 Evolution of Transport and Communication

Topics covered

- Development of transport and trade
- The evolution of communication

Talk about these points:

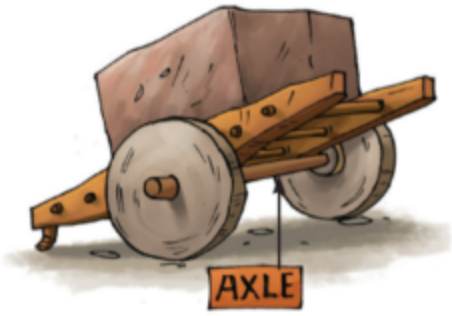
- Which are the places where wheels are used?
- How many can you think of?
- What would our lives be like without the wheel?

A wheel

TP DEVELOPMENT OF TRANSPORT AND TRADE

The wheel

The wheel, a hugely important part of our lives, was invented about 5,000 years ago. The earliest human beings did not have vehicles to carry large and heavy things—they dragged them from place to place. Some people realized that things that roll move faster than things that are dragged. They started moving heavy objects with the help of rollers made of wood. Gradually people invented wheels, which made it much easier to move things from place to place.



Heavy things were moved by using logs of wood as rollers. Gradually wheels with axles were invented.

The earliest vehicles to use the wheel were animal-drawn vehicles, like carts. We know this because of various sources, like the clay cart in this picture.

A 3,000-year-old clay vase in the shape of a wheeled cart

Powering early vehicles

You have learnt that as villages and towns had surplus goods, trade started. Wheeled vehicles, like carts, could transport goods from place to place. The muscle power of bullocks or horses was used to move the vehicles.

People also transported goods by water. Boats used the muscle power of rowers, while sail boats used wind power.

A painting of a trading ship used by traders more than 3,000 years ago—notice the oars and the sail.

Discuss

How is the power used by early human beings different to the power used these days for vehicles?

TP Trade with distant places

At first people traded with places that were not too far away, which they could reach easily. As trade grew, people began to travel to more distant places. This trade happened over land and by sea. As means of transport improved, trade grew.

Carts could not be used for travel over land, as there were few roads. So, for overland trade, goods were carried on animals like camels through the deserts of Asia and Africa. It was easier to transport goods to some places by sea in sailing ships.

Think and answer

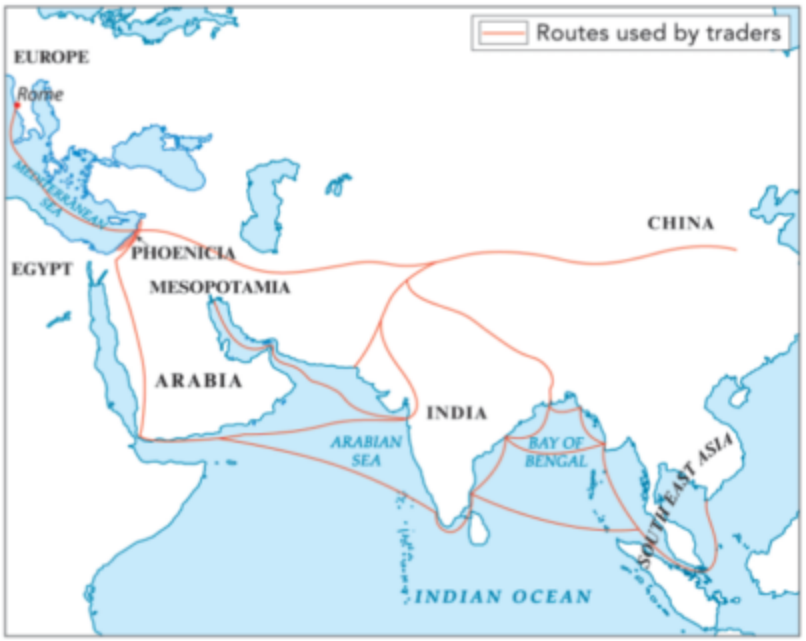
In ancient times, trade journeys could be dangerous and difficult. They also took many months or years since travel was slow. Why do you think trade still continued to grow and develop?

People travelled long distances in the ancient world because of trade. Marco Polo, a trader who lived in the 13th and 14th centuries CE, travelled from Europe over land to China. He wrote a book about his travels. The map on the next page shows some of the routes along which ancient traders travelled.

Some centuries later, there were three famous sea journeys from Europe to distant parts of the world (see map 2).

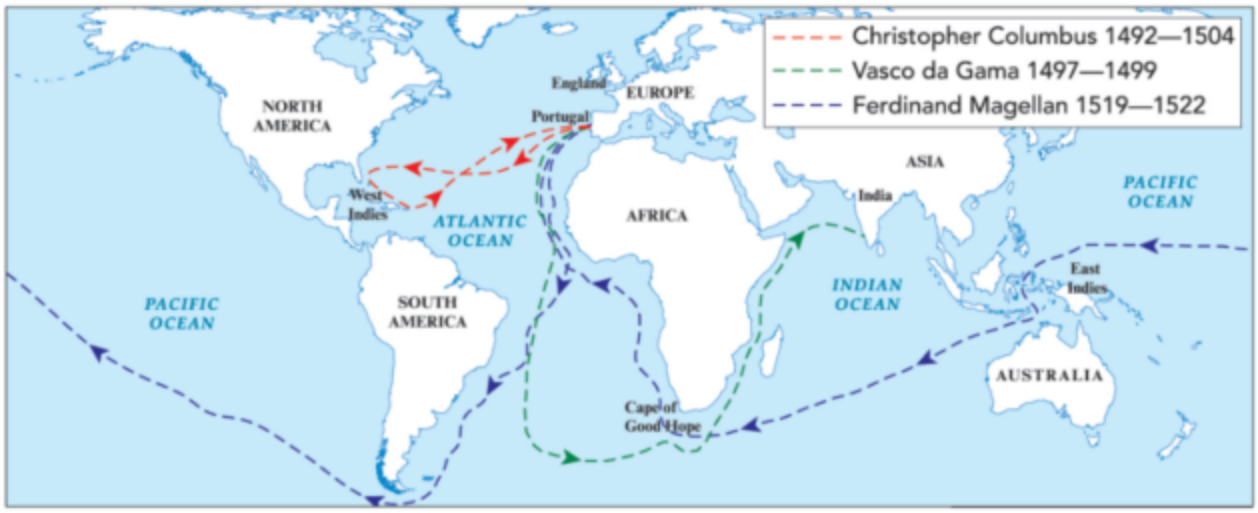
- **Christopher Columbus** tried to find a trade route to India by sea, because the trade route over land had various difficulties. He sailed west from Europe, but reached America instead.
- **Vasco da Gama** also made his journey to show that it was possible to reach India from Europe by sea.
- **Ferdinand Magellan** sailed west from Europe looking for a way to reach the East Indies (South-east Asia), also important for trade. His ships returned to Europe, having sailed around the world. This proved that the earth is round. Until then people had believed that the earth is flat.

The long journeys of these explorers were possible because **shipbuilding** in Europe had improved a great deal.



Map 1: land and sea routes of the ancient world

A replica of one of the ships in which Columbus sailed



Map 2: the journeys of the three explorers

Travelling becomes faster and faster

Steam escapes with great force from a pressure cooker.

For thousands of years human beings used the power of the wind or muscle power for vehicles. In the 18th and 19th centuries, the power of **steam** was discovered, which changed travelling.

New steam ships used this power in engines fitted in their ships. Sea journeys became faster. A journey from England to India, that took several months by sailing ships, could be made in six weeks. Steam was also used to power **locomotives** (engines) of the first trains, which were invented in the early 19th century in England.

Later in the 19th century, crude oil, or **petroleum**, was discovered. Petrol is made from petroleum. Towards the end of the 19th century, the first motor car that used petrol was invented in Germany.

In the early 20th century, the first short flight of an aeroplane using a petrol engine took place in the United States.

Did you Know?

In 1888, Bertha Benz wanted to prove to her husband that the car he had invented could travel long distances. Without telling him, she took the car and travelled over 100 kilometres. The journey took from dawn to dusk and proved that motor car travel was possible.

The first car that could travel long distances

The first flight

Discuss

How were the early car and the aeroplane different from the cars and aeroplanes of today?

During the 20th century, many advances took place in all types of transport. Today **electricity** is also used to power railway engines and some motor cars. Transport today is quite different from early transport, but vehicles that use animal power are still found in some places. The time taken for most journeys is also much less—for example, an air journey from India to England takes only 8-10 hours.

THE EVOLUTION OF COMMUNICATION

Communication means to be in touch with others through speaking, writing, reading or travelling.

We can communicate easily and quickly, even with people who may be far away, because of **technology**. Just as travel and transport were very different in earlier times, so were the methods of communication.

Discuss

How do you communicate with friends and family in the same town or in other places?

Early methods of communication

The earliest people communicated only by speaking. As cities and civilizations developed, people felt the need to write down things like details of their trade, poems, stories, messages, etc. Today people all over the world use writing as an important way of communicating with others.

Did you Know?

The first writing developed in Mesopotamia over 5,000 years ago. People wrote on pieces of clay, called tablets, with 'pens' made of reeds, a type of water plant. Paper was first invented In China several thousand years later.

For hundreds of years, letters were carried by people who travelled by walking, riding, or by carriages or boats. Sometimes letters took many months to reach.

About 3,000 years ago, people realized that **homing pigeons** had the ability to fly back to their homes. They could also be trained to fly between two places. People began to use pigeons to deliver messages quickly. Brief messages were tied to the leg of a pigeon who flew off to deliver it. This was so successful that pigeons were used even during battles, right up to the 20th century, to send urgent messages.

A pigeon with a message tied to its leg

The first printing press

Technological developments in communication

Printing press: All writing was done by hand until the 15th century CE, when the first **printing press** was invented in Germany. Letters of the alphabet, made of wood or metal, were arranged on a machine to form words and sentences. These were inked and sheets of paper were then printed.

Telegraph, telephone and typewriter: In the 19th century, the **telegraph**, a method of sending messages quickly using electricity, was developed both in England and in the United States. Later, also in the 19th century, the first **telephone** conversation took place in the United States. The first **typewriter** was invented at about the same time, also in the United States.

A telephone and typewriter of the early 20th century

Radio and television: Messages were first sent by **radio** in the early 20th century. Later, the first **television** message was transmitted.

A radio of the mid-20th century

Because of these technological developments, communication became much easier and faster than before.

- The printing press made it possible to print many copies of books and newspapers, so knowledge and information became available to many people.
- The telegraph made it possible to send important messages much quicker than sending a letter.
- With the telephone, people who were far away from each other could have a conversation.
- With the typewriter, people could type letters and documents much faster than writing them by hand.
- Radio and television made it possible to deliver news, information and entertainment to thousands of people at the same time. They became very popular all over the world.

Modern communications

In the 19th century, the **postal system** was developed so that people could send letters and parcels by post. Letters posted in a post office were sent by train or by boat and were delivered by postmen. Senders paid for this service by buying stamps. Today, letters are still delivered by the post office, but there are also courier services that deliver letters and packages.

Much of the change in our methods of communication has happened because of developments in technology.

A satellite orbiting the earth

- **Satellites**: The first man-made satellite was sent into space in the mid-20th century. Today such satellites have many different functions. Communication satellites allow radio, television, and telephone **transmissions** to be sent live anywhere in the world.
- The **Internet**: This is a world wide web of connections which makes instant communication possible.
  - With computers, laptops, tablets or smartphones, the Internet is used to:
  - send and receive emails
  - get information from any part of the world
  - use social networking sites
  - upload and download pictures
  - book travel tickets and shop online
- **Mobile phones**: Mobile phone networks are found in many places where the earlier telephone system could not reach. This has helped many more people to be connected.

Did you Know?

The keyboard invented for the typewriter, called 'qwerty', is used today for computers, laptops and smartphones.

Developments in transport and communication make the world seem like a smaller place as we can travel or communicate much faster than before.

Values and Life Skills

The wheel and technology have made transport very comfortable for us, but the fuel used in so many thousands of vehicles pollutes the atmosphere. How can we balance our need for comfortable transport with the need to have clean air?

REMEMBER

1. The invention of the wheel made transport easier.
2. Travel grew because of trade; several famous journeys were made.
3. Inventions in the 19th and 20th centuries made travel easier and faster.
4. In ancient times letters were carried by people, but pigeons were also used sometimes.
5. Communication became easier because of several inventions in the 19th and 20th centuries.
6. Modern communication is very fast because of satellites, the Internet, computers, mobile phones, etc.

NEW WORDS

Locomotive	a railway engine that pulls a train
Technology	use of scientific knowledge, for example in designing new machines
Homing pigeon	a pigeon that has been trained to find its way home from a long distance away
Transmission	sending out signals or messages or broadcasting radio or television programmes

EXERCISES

A. Fill in the blanks.

1. The earliest wheeled vehicles were ..... .
2. As trade expanded in early times, people travelled to more ..... places.
3. Travel by steamships ..... the time taken for journeys.
4. In ancient times, letters sometimes took many ..... to reach.
5. Communications became ..... and ..... because of several technological developments.

B. Write true or false.

Correct the false statements in your notebook.

1. Things that are dragged move as fast as things that are rolled.
2. For many centuries, the muscle power of animals was used for land transport.
3. Many inventions in communication took place in the 19th and 20th centuries.
4. Senders buy postal stamps to pay the postal service for delivering their letters.
5. Satellites do not help to send live radio, television, and telephone transmissions.
6. The Internet is a world wide web of connections which makes instant communication possible.

C. Match the following.

1. The European travellers with their journeys

Marco Polo	his ship sailed around the earth, proving that it is round.
Christopher Columbus	found a sea route from Europe to India.
Vasco da Gama	travelled to China and back.
Ferdinand Magellan	discovered America.

2. The country with the inventions made there

England	telegraph, telephone, typewriter
United States	printing press, motor car
Germany	steam engine, telegraph

D. Choose the correct answer.

1. The wheel was invented about 5,000/2,000 years ago.
2. The power of steam was used for aeroplanes/locomotives.
3. Homing pigeons/Owls were used to send messages quickly during war.
4. The postal/telegraph system was developed to deliver letters and parcels.
5. Smart mobile phones are only useful for telephone calls/can be used as mini-computers.

E. Answer the following questions.

1. How were goods for trade transported in ancient times?
2. What power was used by early ships and how was this different from the power used in ships of the 19th century?
3. How has travel between India and Britain changed in the type of transport and the time the journey takes? Explain in two or three sentences.
4. What does communication mean?
5. Arrange these 'power' words in the right time order: electricity, muscle, petrol, steam
6. Write one sentence each about the advantages of the following inventions—printing press, telegraph, telephone and radio and TV.
7. Give five ways in which the Internet can be used.

THINGS TO DO

TP A. Picture study

This is a page from a book written and decorated by hand. Talk about:

- How long would it have taken to write such books?
- Would they have been expensive?
- Would books have been easily available to people?
- After the printing press was invented, many copies of books became available. What difference would this have made to people?

TP B. In a group

Make a chart or PowerPoint presentation of the various types of transport and communication of earlier times that are no longer used today.

TP C. On your own

In your notebook, arrange these events and their dates in the right order

Transport		Communication	
1885	Karl Benz's petrol automobile	1925	First live TV transmission
1903	First flight at Kitty Hawk	1440	First printing press
1271	Marco Polo starts journey	1876	First telephone conversation
1807	First boat to use steam	1901	First long distance radio signal
1492	Columbus's first journey	1868	First typewriter
1519	Magellan starts journey	1957	First satellite sent into space
1497	Vasco da Gama starts journey	1837	First electric telegraph in England
1829	Steam locomotive called *Rocket*		

WEBLINKS

- Bertha Benz's ride: http://www.uh.edu/engines/epi2402.htm (Accessed on 20 July 2017)
- Here to there—an animated history of transportation: https://www.youtube.com/watch?v=FaLCQo8NJFA (Accessed on 20 July 2017)